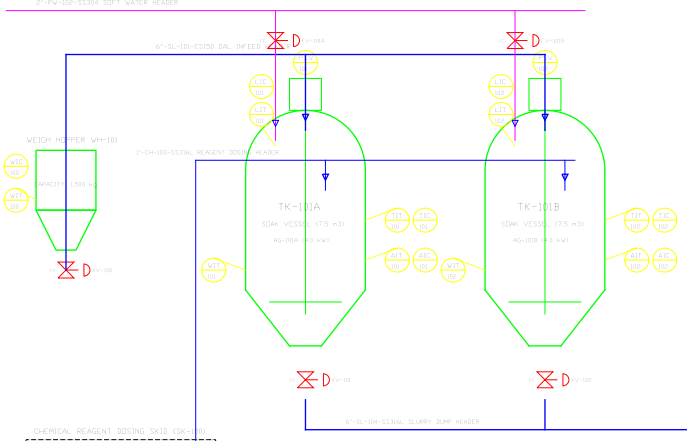

2"-PW-102-SS304 SOFT WATER HEADER
6"-SL-101-CS150 DAL INFEED
WEIGH HOPPER WH-101
1"-CH-103-SS316L REAGENT DOSING HEADER
TK-101A
SOAK VESSEL (7.5 m3)
AG-101A (10 kW)
TK-101B
SOAK VESSEL (7.5 m3)
AG-101B (15 kW)
DEWATERING SCREEN SC-101
1.0 mm WEDGE WIRE BEAR
6"-SL-104-SS316L SLURRY PUMP HEADER
CHEMICAL REAGENT DOSING SKID (SK-101)
TK-102
SODA ASH
P-102A
P-102B
DOSING PUMPS
SUMP TK-103
P-103 (ETP PUMP)
CONTROL LOOPS, INTERLOCKS & SAFETY PHILOSOPHY (ISA-84 / SIL-1)
1. LIC-101/102: Raise level control. High-High level (HHS) triggers I-01 tripping conveyor CV-101.
2. AIC-101/102: Inline pH analytical control modulates P-102A/B stroke speed to maintain target pH 8.2–8.4.
3. TIC-101/102: Temperature loop modulates FV-101 hot/cold blending to maintain soak water at 38 deg C +/- 1 deg C.
4. XV-101A/B Fail-Closed pneumatic dump valves interlocked with Dewatering Screen SC-101 running status.
5. ESS-101 Emergency Shutdown isolates all chemical dosing pumps and initiates emergency CIP flush.
HARAKUTIR AGRO-PROCESS ENGINEERING PVT. LTD.
15,000 kg/day Spirit Pigeon Pea Optimization Greenfield Project
P&ID – SOAKING BATTERY & CHEMICAL DOSING
PROCESSING PERMIT / STATE CONDITIONAL FEED
DWG NO: DWG-FIN-IST-003
REV: V4.0
SHEET: 3 OF 14
SCALE: AS SHOWN
DATE: 2026-11-05
STATUS: PRELIMINARY
DATE: 04 FEB 2026
CONCEPTUAL FEED – PRELIMINARY – NOT FOR CONSTRUCTION / NOT FOR PROCUREMENT